

Association Rule Learning

on the Groceries Dataset

Unsupervised Learning Methods – Assignment

Association Rule Learning

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Q1. Understand the Data

(a) Number of transactions

The Groceries dataset contains **9,835 transactions**, each representing one customer's market basket.

(b) Number of unique items

The dataset contains **169 unique grocery items**.

(c) Ten most frequently purchased items

Table 1 and Figure 1 show the ten items with the highest support (relative purchase frequency).

Table 1: Top 10 most frequent items

Item	Support	Count
Whole milk	0.2555	2,513
Other vegetables	0.1935	1,903
Rolls/buns	0.1839	1,809
Soda	0.1744	1,715
Yogurt	0.1395	1,372
Bottled water	0.1105	1,087
Root vegetables	0.1090	1,072
Tropical fruit	0.1049	1,032
Shopping bags	0.0985	969
Sausage	0.0940	924

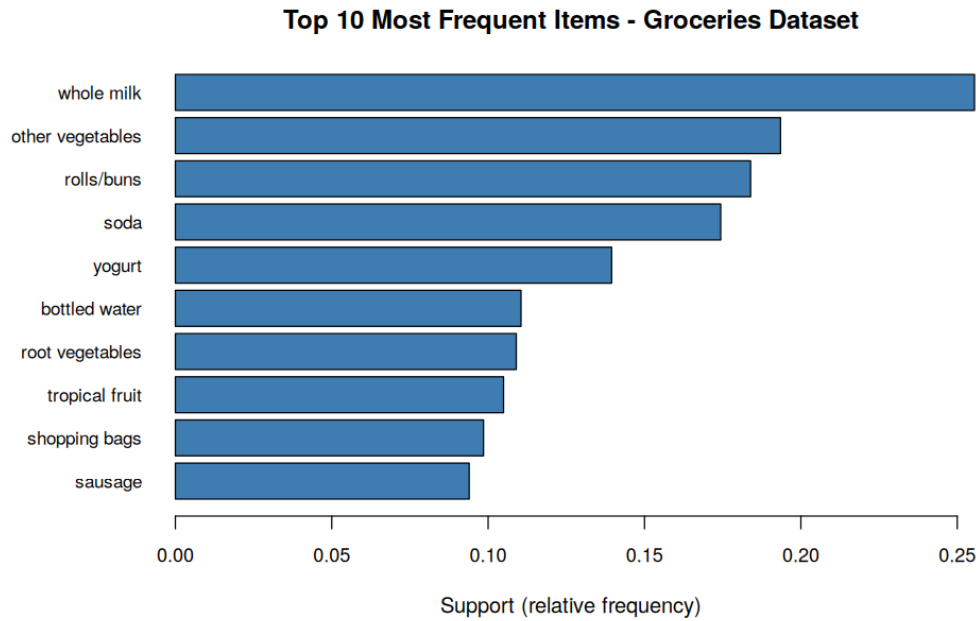


Figure 1: Top 10 most frequent items by support

Whole milk and other vegetables are, by a clear margin, the two staples of this store's customer base – unsurprising, since both are everyday perishables bought on nearly every shopping trip.

Q2. Frequent Itemsets

(a) At least 10 frequent itemsets (two or more products)

Using a minimum support threshold of **0.02** (i.e. the itemset appears in at least 2% of all transactions) and a minimum itemset size of 2, the analysis identified **63 frequent itemsets** (61 pairs and 2 triples). Table 2 lists the top 10 by support.

Table 2: Top 10 frequent itemsets by support

Itemset	Support	Count
{other vegetables, whole milk}	0.0748	736
{rolls/buns, whole milk}	0.0566	557
{whole milk, yogurt}	0.0560	551
{root vegetables, whole milk}	0.0489	481
{other vegetables, root vegetables}	0.0474	466
{other vegetables, yogurt}	0.0434	427
{other vegetables, rolls/buns}	0.0426	419
{tropical fruit, whole milk}	0.0423	416
{soda, whole milk}	0.0401	394
{rolls/buns, soda}	0.0383	377

(b) Support values

Support values are reported directly in Table 2 (column *Support*), ranging from 0.0383 to 0.0748 for the top 10 itemsets.

(c) Highest-support itemset

{other vegetables, whole milk} has the highest support at **0.0748** (736 of 9,835 transactions) – almost twice the support of the next-highest itemset.

(d) What support tells us about purchasing behavior

Support measures how common an itemset is across *all* transactions, regardless of what else is happening in the basket. A support of 0.0748 for {other vegetables, whole milk} means that roughly 1 in 13 shopping trips includes both items together. High-support itemsets identify the combinations that matter at scale – store layout, staffing, and inventory decisions built around them affect a large share of all customers, whereas a rule with strong confidence but tiny support only describes a small, possibly niche, slice of shoppers.

Q3. Association Rules

Measures used

Three measures are used throughout this report, for a rule $A \Rightarrow B$ and an itemset X (with T the set of all transactions):

$$\text{support}(X) = \frac{|\{t \in T : X \subseteq t\}|}{|T|} \quad \text{confidence}(A \Rightarrow B) = \frac{\text{support}(A \cup B)}{\text{support}(A)} \quad \text{lift}(A \Rightarrow B) = \frac{\text{confidence}(A \Rightarrow B)}{\text{support}(B)}$$

(a) At least 10 association rules

Rules were generated from the frequent itemsets above using a minimum confidence of **0.25**, yielding **49 rules** in total. Table 3 lists the top 12, sorted by lift.

Table 3: Association rules (sorted by lift)

Antecedent	Consequent	Support	Confidence	Lift
other vegetables, whole milk	root vegetables	0.0232	0.3098	2.8421
pip fruit	tropical fruit	0.0204	0.2702	2.5746
root vegetables, whole milk	other vegetables	0.0232	0.4740	2.4498
root vegetables	other vegetables	0.0474	0.4347	2.2466
other vegetables, whole milk	yogurt	0.0223	0.2976	2.1330
whipped/sour cream	other vegetables	0.0289	0.4028	2.0819
whipped/sour cream	yogurt	0.0207	0.2894	2.0743
whole milk, yogurt	other vegetables	0.0223	0.3975	2.0541
other vegetables, yogurt	whole milk	0.0223	0.5129	2.0072
tropical fruit	yogurt	0.0293	0.2791	2.0005
butter	whole milk	0.0276	0.4972	1.9461
pork	other vegetables	0.0217	0.3757	1.9415

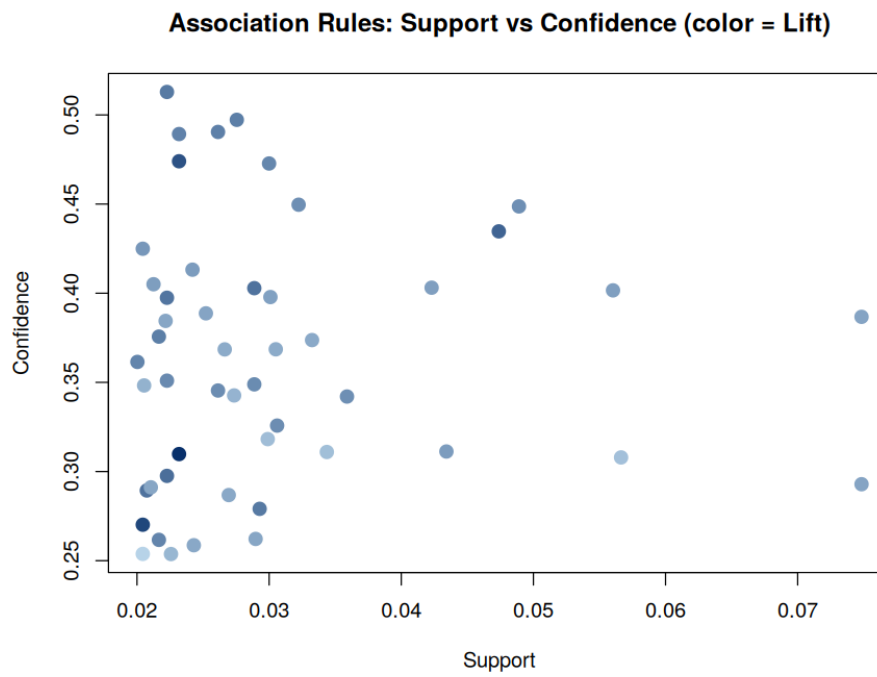


Figure 2: All 49 rules: support vs. confidence, coloured by lift (darker = higher lift)

Q4. Understanding the Measures

Three rules from Table 3 are examined below.

(a) Support

- **{other vegetables, whole milk} → {root vegetables}**:

$$\text{support} = \frac{|\{t : \{\text{o.veg, milk, root veg}\} \subseteq t\}|}{9,835} = \frac{228}{9,835} = 0.0232$$

this exact combination of all three items occurs in 2.32% of all transactions.

- **{root vegetables} → {other vegetables}**: support = 0.0474 – both items together appear in 4.74% of transactions.
- **{butter} → {whole milk}**: support = 0.0276 – both items appear together in 2.76% of transactions.

(b) Confidence

- **{other vegetables, whole milk} → {root vegetables}**:

$$\text{confidence} = \frac{\text{support}(A \cup B)}{\text{support}(A)} = \frac{0.0232}{0.0748} = 0.3098$$

of the customers who bought other vegetables *and* whole milk, 30.98% also bought root vegetables.

- **{root vegetables} → {other vegetables}**: confidence = 0.4347 – 43.47% of customers who bought root vegetables also bought other vegetables.
- **{butter} → {whole milk}**: confidence = 0.4972 – almost half (49.72%) of butter buyers also bought whole milk.

(c) Lift

- **{other vegetables, whole milk} → {root vegetables}**:

$$\text{lift} = \frac{\text{confidence}(A \Rightarrow B)}{\text{support}(B)} = \frac{0.3098}{0.1090} = 2.8421$$

customers who buy other vegetables and whole milk are 2.84 times more likely to also buy root vegetables than a random customer.

- **{root vegetables} → {other vegetables}**: lift = 2.25 – buying root vegetables makes buying other vegetables 2.25 times more likely than chance.
- **{butter} → {whole milk}**: lift = 1.95 – buying butter makes buying whole milk about 1.95 times more likely than chance, despite whole milk already being the most popular item in the store.

Q5. Find Interesting Rules

(a) Highest confidence

$\{\text{other vegetables, yogurt}\} \rightarrow \{\text{whole milk}\}$, confidence = 0.5129 (support = 0.0223, lift = 2.01).

(b) Highest lift

$\{\text{other vegetables, whole milk}\} \rightarrow \{\text{root vegetables}\}$, lift = 2.84 (support = 0.0232, confidence = 0.31).

(c) Highest support

$\{\text{other vegetables}\} \rightarrow \{\text{whole milk}\}$, support = 0.0748 (confidence = 0.3868, lift = 1.51).

(d) Are these the same rule?

No – all three are different rules. This is expected: support rewards *popularity* (the highest-support rule involves the two most common items in the store, so it occurs often even though the association is only moderately strong – lift 1.51 is the lowest of the three). Confidence rewards *reliability* of the antecedent predicting the consequent, which is why the highest-confidence rule points at whole milk, the single most-bought item overall – it’s an easy consequent to satisfy. Lift rewards *genuine association strength* after removing the effect of overall popularity, which is why the highest-lift rule involves root vegetables, a less common item whose purchase is still strongly tied to a specific combination of other items. A rule can score well on one measure and only moderately on the others, since each measure is answering a different question.

Q6. Interpret a Rule

Selected rule: **{other vegetables, whole milk} → {root vegetables}** (support = 0.0232, confidence = 0.31, lift = 2.84).

In simple language: Customers who purchase other vegetables and whole milk in the same trip also tend to purchase root vegetables in about 31% of those trips – noticeably more often than root vegetables are bought overall (support 0.109), which is exactly what the lift of 2.84 captures.

Strong or weak association? This rule represents a **moderately strong** association. The lift of 2.84 is comfortably above 1, meaning the three items co-occur far more than chance would predict, and 31% confidence means nearly a third of the relevant customers follow the pattern. It is not an overwhelming rule (confidence is well under 50%), but combined with a lift close to 3, it is one of the more meaningful relationships found in this dataset – plausibly reflecting a shared ”cooking a vegetable-based meal” shopping occasion.

Q7. Business Application

Imagine acting as the supermarket manager, using the rules above.

(a) Two products to place close together

Root vegetables and other vegetables. Their rule ($\{\text{root vegetables}\} \rightarrow \{\text{other vegetables}\}$, lift = 2.25) shows a strong, high-confidence association (43.5%), and both are fresh produce that is naturally merchandised in the same aisle already – placing them adjacent (or cross-merchandising root vegetables at the front of the vegetable section) would shorten the trip for a large share of customers.

(b) One possible product bundle

A **”vegetable basket” bundle of other vegetables + whole milk + root vegetables** (the highest-lift itemset, lift 2.84) – e.g. a discounted ”dinner veg box” combining these three – targets an already-existing shopping pattern rather than trying to create a new one.

(c) Using association rules for recommendations

At checkout (in-store) or in an online cart, the system could flag: ”Customers who bought other vegetables and whole milk also bought root vegetables (frequently)” and offer a small discount or a reminder nudge – a ”you might also need” prompt driven directly by the rule $\{\text{other vegetables, whole milk}\} \rightarrow \{\text{root vegetables}\}$, rather than a generic or manually curated suggestion.

Q8. Understanding Association

(a) Does an association rule prove causation?

No. An association rule only describes **co-occurrence** in the transaction data – that customers who bought *A* frequently also bought *B*. It says nothing about *why*: *A* might not cause anyone to buy *B* at all. Both could be driven by a common cause (e.g. both are ingredients for the same popular recipe), by seasonality, by store layout placing them near each other, or by pure coincidence in a large dataset. Confirming causation would require a controlled experiment (e.g. moving product placement and measuring the effect), not just mining historical baskets.

(b) High confidence but low lift

This happens when the consequent is already very common on its own. Confidence only asks "given the antecedent, how often does the consequent follow?" – if the consequent (say, whole milk, support 0.2555) is bought in a quarter of *all* transactions regardless of anything else, then almost any antecedent will show reasonably high confidence for predicting it, purely because it's hard to miss. Lift corrects for this by dividing by the consequent's baseline popularity, so a rule can have confidence well above the dataset average while still having lift close to 1 – meaning the antecedent barely changes the odds of the consequent compared to knowing nothing at all.

(c) What a lift greater than 1 indicates

Lift > 1 indicates a **positive association**: the two itemsets occur together more often than would be expected if they were purchased independently. The further above 1, the stronger the association (lift = 1 means no association / statistical independence, and lift < 1 indicates a negative association, where buying one item makes the other *less* likely).